SURP 2026 · MIDTERM PROGRESS REPORT

PCA-RESIDUAL STATISTICAL ARBITRAGE

on Indian Equities & Futures

Kartikya
3rd-Year, Industrial Engineering & Operations Research · IIT Bombay
Team A — Strategy Design | Guide: Prof. Sandip Mondal (EE, IIT Bombay)

OVERVIEW

THE PROJECT AT A GLANCE

THE GOAL

Build & honestly evaluate a market-neutral statistical-arbitrage strategy for Indian markets — not just chase a good-looking backtest.

THE METHOD

PCA strips common market & sector factors from returns; we trade the mean-reversion of the leftover stock-specific residual.

THE DISCIPLINE

Realistic Indian costs, out-of-sample tests, known-answer validation — every result independently cross-checked.

THE ARC

Cash equities studied end-to-end (complete). Now extending the identical strategy to single-stock futures.

THE STRATEGY

HOW PCA-RESIDUAL REVERSION WORKS

1	**DECOMPOSE**	Run PCA on ~200 liquid stocks' daily returns to extract common factors (market, sectors) — the forces all stocks share.
2	**RESIDUAL**	Subtract the factor-explained part from each stock's return. What remains is its own idiosyncratic move.
3	**SIGNAL (S-SCORE)**	Standardize the residual: how many std-devs from normal is this stock? High = 'rich', low = 'cheap'.
4	**TRADE**	Short the rich ($s > +2$), long the cheap ($s < -2$), dollar-neutral. Bet on relative reversion, not market direction.
5	**SIZE & EXIT**	Inverse-volatility sizing, gross capped ~100%. Triple-barrier exit: revert (profit), diverge (stop), or 15-day time-stop.

Faithful implementation of the Avellaneda–Lee (2010) framework — parameters kept at literature defaults, deliberately not tuned.

PART I

THE CASH-EQUITY STUDY

10 years of real NSE data · 435 stocks · 2015–2024 · fully realistic Indian costs



PART I · SETUP

DATA & EXPERIMENTAL SETUP

2,466 trading days (2015–2024)

435 NSE stocks (cleaned universe)

~200 traded daily, ranked by liquidity

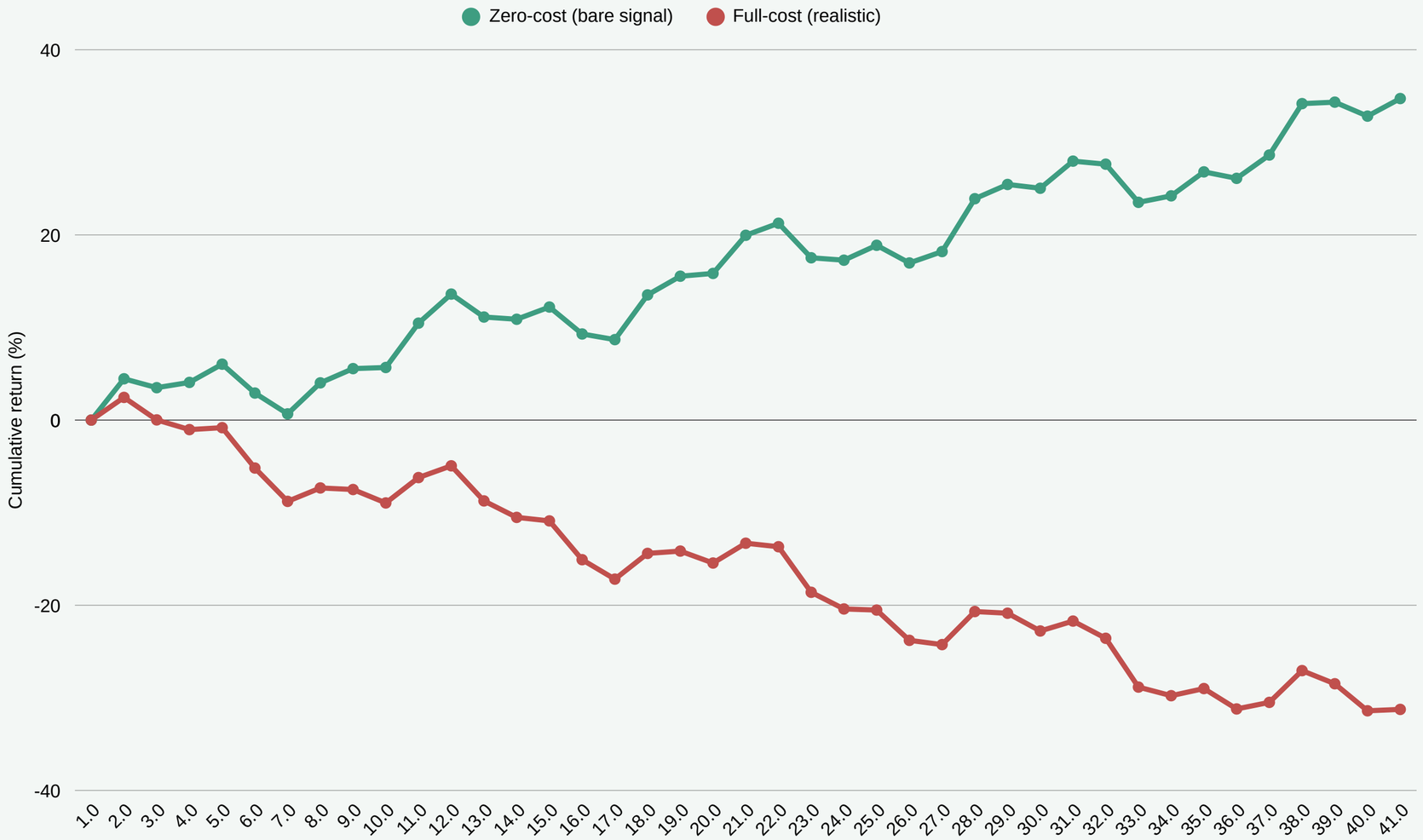
₹10 L starting capital · ~100% gross cap

BUILT FOR REALISM

- Adjusted OHLCV via yfinance — splits & dividends handled
- Look-ahead-safe: signal at T close → trade T+1 open → mark T+1 close
- Realistic Indian cost model: STT + slippage + brokerage
- Late IPOs excluded until listed — no phantom trades
- Full Python pipeline: signal → decision → execution → P&L

PART I · HEADLINE RESULT

THE SIGNAL HAS EDGE — COSTS ERASE IT



+32.9%

Zero-cost return

Sharpe +0.46

-32.9%

Full-cost return

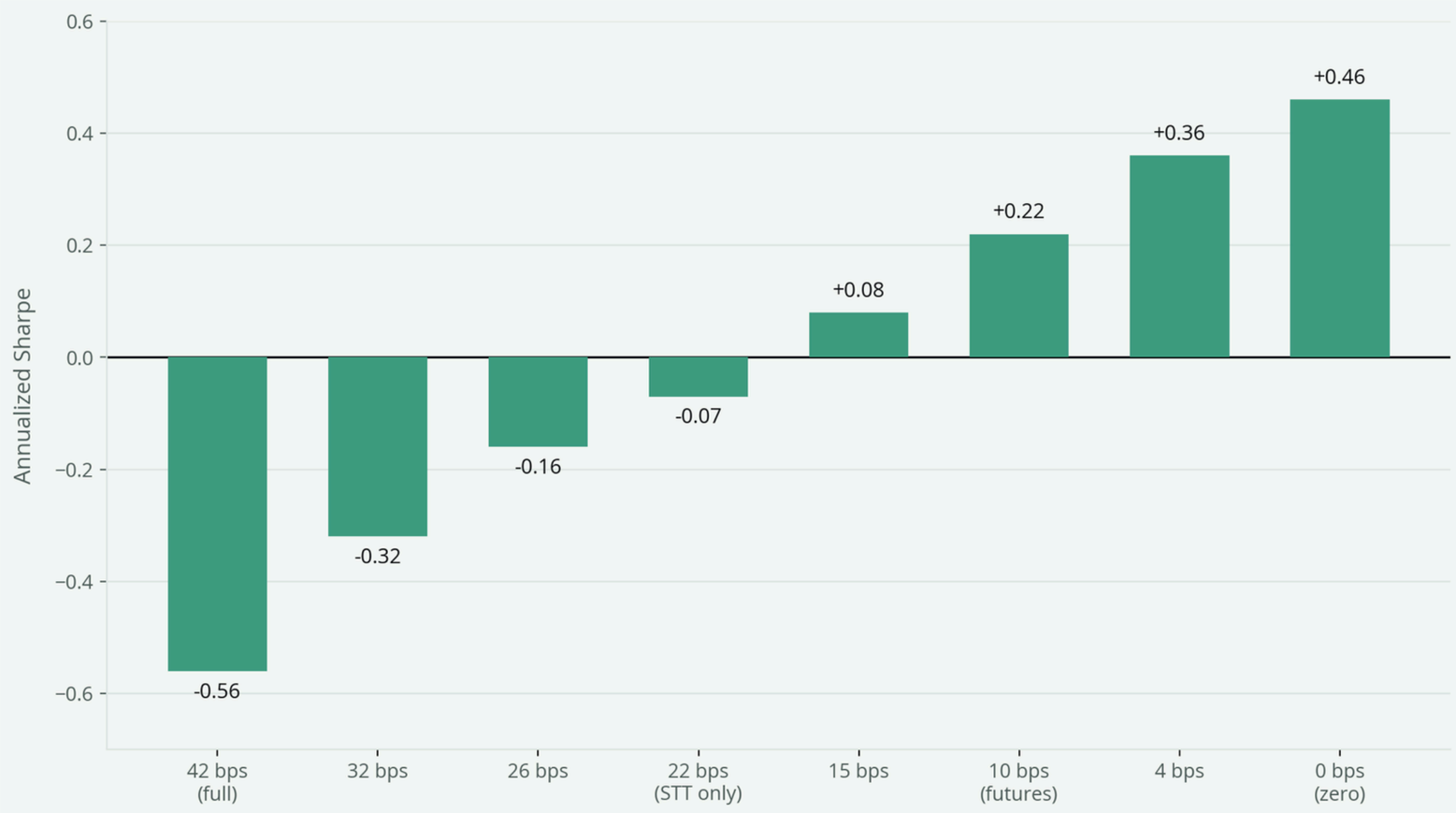
Sharpe −0.56

₹6.15 L

Total costs paid

≈ 2× the gross edge

WHY IT FAILS: THE TAX ALONE BEATS THE EDGE



THE STRUCTURAL WALL

Break-even ≈ 18–20 bps

India's STT floor alone is ~22 bps round-trip — a government tax you cannot avoid.

At 'STT only' (zero slippage), the strategy is already negative.

→ Untradable in cash equities under any execution — positive only in a low-cost futures regime.

PART I · STATISTICAL VALIDITY

IS THE REVERSION REAL? OOS SAYS YES

We froze the PCA factors on past data and tested whether today's s-score predicts FUTURE residual returns (Fama–MacBeth). Over 10 years the effect is small but statistically significant at every horizon.

Horizon	Correlation	t-statistic	Hit rate	Significant?
1 day	−0.019	−2.35	54.1%	Yes ✓
3 days	−0.021	−2.75	55.1%	Yes ✓
5 days	−0.019	−2.25	54.9%	Yes ✓
10 days	−0.028	−2.71	55.9%	Yes ✓

WHAT THIS MEANS

|t| > 1.96 = statistically real
The reversion genuinely exists on NSE equities — not noise.
But correlation ~0.02 is tiny — edge per trade is minuscule.
Statistically significant, economically too thin to beat costs.

PART II

THE FUTURES EXTENSION

Same strategy, lower-cost regime — where the cash study says an edge could survive



PART II · MOTIVATION

WHY SINGLE-STOCK FUTURES ARE THE PIVOT

COSTS ~4× LOWER

No STT-scale tax (~0.02% sell-side only). All-in ~10–15 bps vs ~42 bps in cash — exactly the regime where the edge turns positive (Sharpe ~+0.22).

SHORTING IS REALISTIC

The cash study assumed free overnight shorting — not truly possible for retail. Futures short freely overnight, making the market-neutral book genuinely executable.

FREE 10-YEAR DATA EXISTS

NSE F&O bhavcopy — official daily archive, free, no account. Covers all single-stock & index futures 2015–2024, including contract-roll construction.

MINIMAL CHANGE TO METHOD

The identical PCA-residual pipeline runs on a continuous futures series. We reuse the verified engine — only the instrument and cost model change.

KEY TAKEAWAYS

WHERE WE STAND

A complete, rigorous result

PCA-residual reversion on NSE cash equities is statistically real but economically untradable — the STT floor alone exceeds the edge. Triple-verified, honestly reported.

Research-grade discipline

Out-of-sample Fama–MacBeth testing, known-answer validation, a caught P&L bug and two avoided overfitting traps — the negative result is credible, not a dead end.

A clear, motivated next step

The cost analysis points precisely to futures. The free data pipeline is already verified end-to-end; the definitive futures backtest is the immediate goal.

THANK YOU · Kartikya · Guide: Prof. Sandip Mondal